TEXORA · SESSIONAL NOTES · STUDY SHEET

Trouser Making with Industrial Sewing Machine

The complete making sequence of the basic trouser on industrial machines: pockets, zip fly, leg joining, waistband setting, closure and hemming, with the quality gates between operations.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 8 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to make a complete basic trouser (size M) on industrial machines from the Studies 3–5 patterns: preparing parts, constructing pockets and fly, joining legs, setting the waistband, hemming and finishing, in the factory operation sequence.

2. Materials, machine settings and specimens

Materials and machines: cut parts from the Studies 4–5 patterns (shell, pocketing, interlining); SNLS machine (301, Nm 90, core-spun tex 40), 5-thread overlock (516/504), bartack machine, buttonhole and button-attach or hook setting, blind-stitch or SNLS for the hem. **Settings:** density 4–4.5 stitches/cm on wovens; serging before assembly where the sequence requires; fusing of the waistband interlining at the Topic 7 tested conditions before the band is closed.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. Operation sequence

STEP	OPERATION	MACHINE / STITCH AND QUALITY POINT

1	Serge leg pieces; make darts (back)	504 on edges; dart sewn 301, pressed toward centre back.
2	Front pockets: attach bag + facing, close bag	301; opening topstitched; bag closed and bartacked at both ends of the opening.
3	Fly: set zip between fly piece and shield	301; zip teeth covered by the fly; J-stitch topstitch on the fly line; bartack at the fly base.
4	Join inseams, then outseams (side seams)	301 or 516; notches matched at knee; allowances 1.0 inseam / 1.5 side per the pattern.
5	Join the crutch (seat) seam through	301 × 2 rows or 401; the two inseams aligned at the fork without a step.
6	Waistband: prepare (fuse, fold), set to waist, close ends, topstitch	301; band length check against waist 82 before closing; notches at centre back and side seams.
7	Belt loops (if styled): make and set	Folder + 301; bartack both ends.
8	Buttonhole + button (or hook-and-bar); hem the bottoms	Buttonhole sized to the 24L (15.2 mm) button; hem 2.5 cm turned, blind stitch or 301 per spec.
9	Finish: trim threads, press (Topic 8), measure	Full POM check against the chart: waist 82 ± 1, inseam 78 ± 1, hem 40 ± 0.7.

4. Critical quality points during making

- After step 3: fly lies flat, zip opens fully, shield covers the teeth from inside; the J-stitch is even (about 3.5 cm wide at its curve).
- After step 5: fork junction has no step or hole; seat seam lies smoothly when the trouser is folded on its crease lines.
- After step 6: waist measures 82 before the closure is set — a band sewn to a wrong waist cannot be corrected at the button.
- Both legs equal: inseams compared leg-to-leg before hemming; hems level when hung by the waistband.

Observation points — what to record

- Time or difficulty at each step, honestly recorded — the record is a method-study document.

- Fly quality after step 3 (flatness, J-stitch width, zip travel).
- Final POM measurements of your garment against the chart, with pass/fail per point.
- Any fault and its Topic 4 classification (formation, appearance, damage) with the remedy applied.

Precautions

- Measure the waist before setting the closure (step 8) — after it, a waist fault is a remake.
- Press pockets, fly and darts as you go (under-pressing, TE-3107 Topic 8); a finished trouser cannot be fully corrected at top pressing.
- Bartack the pocket openings and fly base before the garment leaves the line section — unbartacked stress points fail in wear, not in the lab.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why are the leg edges serged before assembly in this sequence, and on which later operations would serging be impossible?
2. State the final POM check values for waist and inseam with tolerances.
3. A finished trouser's fly gapes at the base although the zip works. Which step and which omitted reinforcement explain it?
4. In what order are inseam and outseam joined here, and why must the crutch seam come after both?

Answer key

1. Serging (504) neatens single-ply edges; once seams are closed and the waistband set, the overlock cannot reach the inner allowances of the legs and fly area. Edge finishing is therefore scheduled while the parts are flat.
2. Waist 82 ± 1.0 cm; inseam 78 ± 1.0 cm, measured as finished garment POMs (TE-2207 system).
3. Step 3 without its closing bartack at the fly base (and/or the J-stitch ending short): the fly's lower end is the stress point of the closure; without the tack it opens under wear load.
4. Inseams first, then outseams, then the crutch through in one continuous seam: the seat seam joins two already-closed legs, and sewing it earlier would leave the legs unjoinable in the flat. The fork junction must align the two inseams exactly.

5. Method note: why this order is the factory order

The sequence of §3 is not arbitrary; it follows three rules that govern all garment assembly. **Flat before closed:** every operation that can be done on a flat, single part (serging, darts, pockets, fly) is done before the part is joined into a tube, because flat work is faster, visible and correctable. **Small before large:** sub-assemblies (pocket units, the waistband unit, loops) are completed as units before they meet the main garment. **Stress points last-checked, first-protected:** bartacks are placed while the garment is still in the section that made the joint, not left to a final repair station. A student who can state, for any two adjacent steps, why their order cannot be swapped understands the sequence; one who has only memorised the list does not — which is why the viva questions below probe the order itself.
